

Quadratic forms and optimization

- Review:
 - a) Symmetric matrices: $A^T = A$.
 - b) Fundamental fact about transpose: $\mathbf{u} \cdot A\mathbf{v} = (A^T\mathbf{u}) \cdot \mathbf{v}$.
 - c) If A is symmetric,
 - 1) Two eigenvectors with distinct eigenvalues are orthogonal.
 - 2) All eigenvalues are real.
 - d) Definition: An $n \times n$ matrix A is said to be *orthogonally diagonalizable* if there is a diagonal matrix D and orthogonal matrix P with $P^T = P^{-1}$ such that $A = PDP^{-1}$.
 - e) Theorem: A matrix is orthogonally diagonalizable if and only if it is symmetric.
 - 1) Thus, A has n real eigenvalues counting multiplicity.
 - 2) The eigenvectors form a complete, orthogonal basis.
 - f) Spectral theorem: Let A be a symmetric matrix. Fix an orthonormal basis $\{\mathbf{v}_k\}$ of eigenvectors. Then $A = \lambda_1 \mathbf{v}_1 \mathbf{v}_1^T + \cdots + \lambda_n \mathbf{v}_n \mathbf{v}_n^T$.

Definition: A quadratic form on $\mathbb{R}^n$ is a function (to be denoted by Q) whose value at $\mathbf{x}$ can be computed using the rule $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x} = \mathbf{x} \cdot A \mathbf{x}$ where A is a symmetric matrix.

1. Which of the following functions are quadratic forms (and which aren't)? For those that are, write the corresponding symmetric matrix.
 - a) $Q(\mathbf{x}) = \mathbf{x} \cdot \mathbf{x}$
 - b) $Q(\mathbf{x}) = -x_1^2 - x_2^2 + x_3^2 + x_4^2$.
 - c) $Q(\mathbf{x}) = x_1 x_2 + x_3 x_4$
 - d) $Q(\mathbf{x}) = x_1 - x_2 x_3$.
 - e) $Q(\mathbf{x}) = \cos(x_1^2 - x_2^2)$.

2. Consider the quadratic form $Q(\mathbf{x}) = x_1^2 - x_2^2$. Compute the quadratic form $\mathbf{y} \rightarrow Q(P\mathbf{y})$ where P is the matrix below:

a) $P = \begin{pmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$.

b) $P = \frac{2}{\sqrt{15}} \begin{pmatrix} 2 & \frac{1}{2} \\ \frac{1}{2} & 2 \end{pmatrix}$

Principle axis theorem: Given a quadratic form $Q(\mathbf{x})$, there is a change of variables $\mathbf{x} = P\mathbf{y}$ with P orthogonal such that the quadratic form $\mathbf{y} \rightarrow Q(P\mathbf{y})$ is $\mathbf{y}^T D \mathbf{y}$ with D being diagonal (the eigenvalues of A are its diagonal entries).

3. Diagonalize the following quadratic forms (find the diagonal matrix D and the orthogonal change of variable matrix P):

a) $Q(\mathbf{x}) = x_1^2 - 2x_1x_2 + x_2^2$.

b) $Q(\mathbf{x}) = 2x_1^2 - 6x_1x_2 + 2x_2^2$.

c) $Q(\mathbf{x}) = x_1x_2 + x_3x_4$.

4. For each of the quadratic forms below, what are the minimum and maximum values on the set where $\|\mathbf{x}\|=1$? And at what points are they achieved? And what are the saddle points and the corresponding value for Q ?

a) $Q(\mathbf{x}) = x_1^2 - 4x_2^2$.

b) $Q(\mathbf{x}) = 7x_1^2 + 4x_2^2 + 3x_3^2$.

c) $Q(\mathbf{x}) = -7x_1^2 + 4x_2^2 + 3x_3^2$.

d) $Q(\mathbf{x}) = -7x_1^2 - 4x_2^2 + 3x_3^2$.

e) $Q(\mathbf{x}) = -7x_1^2 - 4x_2^2 - 3x_3^2$.

5. Find the minimum and maximum values of the following quadratic forms on the set where $\|\mathbf{x}\|=1$; and give their coordinates. Also find all saddle points with their coordinates.

a) $Q(\mathbf{x}) = 2x_1x_2 + 2x_2x_3 + 2x_3x_1$.

b) $Q(\mathbf{x}) = 2x_1x_2 - 2x_2x_3$.

c) $Q(\mathbf{x}) = 4x_1x_2 - x_1^2 - x_2^2$.

Quadratic forms and optimization – Answers / Solutions

1. a) Yes, $A = I_n$. b) Yes, $A = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$. c) Yes, $A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$.

d) No. e) No.

2. a) $Q(\mathbf{Py}) = 4y_1^2 - \frac{1}{4}y_2^2$.
 b) $Q(\mathbf{Py}) = y_1^2 - y_2^2$.

3. a) $D = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$, $P = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$.

b) $D = \begin{pmatrix} 5 & 0 \\ 0 & -1 \end{pmatrix}$, $P = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$.

c) $D = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$, $P = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 \end{pmatrix}$.

4. a) Maximum value = 1 where $x_2 = 0$, minimum = -4 where $x_1 = 0$.
 b) Maximum value = 7 where $x_2 = x_3 = 0$, minimum value = 3 where $x_1 = x_2 = 0$, saddle points where $x_1 = x_3 = 0$ with value 4.
 c) Maximum value = 4 where $x_1 = x_3 = 0$, minimum value = -7 where $x_2 = x_3 = 0$, saddle points where $x_1 = x_2 = 0$ with value 3.
 d) Maximum value = 3 where $x_1 = x_2 = 0$, minimum value = -7 where $x_2 = x_3 = 0$, saddle points where $x_1 = x_2 = 0$ with value -4.
 b) Maximum value = -3 where $x_1 = x_2 = 0$, minimum value = -7 where $x_2 = x_3 = 0$, saddle points where $x_1 = x_3 = 0$ with value -4.

5. a) The matrix for A is $\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ which has eigenvalues $\lambda = 2$ with eigenvector

$$\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \text{ and } \lambda = -1 \text{ with a 2-dimensional eigenspace spanned by } \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \text{ and } \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}.$$

Thus, the maximum of Q on the sphere where $\|\mathbf{x}\| = 1$ is 2 at the points $\pm \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$,

and the minimum is -1 on the circle of radius 1 in the plane where $x + y + z = 0$.

b) The matrix for A is $\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}$ which has eigenvalues $\lambda = 0$ and $\lambda = \pm\sqrt{2}$. The

maximum of Q is therefore $\sqrt{2}$ and that is taken on at the points $\pm \frac{1}{2} \begin{pmatrix} 1 \\ \sqrt{2} \\ -1 \end{pmatrix}$, the

minimum is $-\sqrt{2}$ which is taken on at $\pm \frac{1}{2} \begin{pmatrix} 1 \\ -\sqrt{2} \\ -1 \end{pmatrix}$, and then there are saddle

points at $\pm \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ where Q is zero.

c) The matrix A is $\begin{pmatrix} -1 & 2 \\ 2 & -1 \end{pmatrix}$ which has eigenvalues $\lambda = 1$ and $\lambda = -3$. Thus, the

maximum of Q is 1 which occurs at the points $\pm \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, and the minimum of

Q is -3 which occurs at $\pm \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.